







4. The US government monitors the consumption of different products. The table shows  $y$ , the amount of ice cream consumed in millions of pounds, for  $x$  years since 2010.

|     |       |       |       |       |       |       |       |       |       |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $x$ | 0     | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     |
| $y$ | 4,354 | 4,222 | 4,114 | 4,030 | 3,970 | 3,934 | 3,922 | 3,934 | 3,970 |

Write the quadratic equation, in any form, for the quadratic function that models the amount of ice cream consumed, in millions of pounds, since 2010.

|                 |                                                                               |
|-----------------|-------------------------------------------------------------------------------|
| <b>Work</b>     | $y = a(x - 6)^2 + 3922$ $3934 = a(7 - 6)^2 + 3922$ $3934 = a + 3922$ $12 = a$ |
| <b>Equation</b> | $y = 12(x - 6)^2 + 3922 \text{ or } y = 12x^2 - 144x + 4354$                  |